Nexus AI — Architecture & Codebase Guide

Comprehensive Technical Documentation: Laravel 12 + Google Gemini + MySQL + Vision & AI Image Generation

Application: Nexus AI Studio	**Framework:** Laravel 12 (PHP 8.3)	**AI Engine:** Google Gemini API
Database: MySQL (CHATBOT)	**Image Engine:** Pollinations.ai FLUX / SDXL	**Protocol:** Server-Sent Events (SSE)

1. System Overview & Architecture

Nexus AI is an enterprise-grade AI chatbot web application built on **Laravel 12 (PHP 8.3)** and integrated with **Google Gemini's official Generative Language API**. It replaces legacy single-turn chatbots with a fully reactive, multimodal conversational workspace featuring live token-by-token Server-Sent Events (SSE) streaming, persistent MySQL conversation management, multimodal file/image analysis, and autonomous AI image generation.

The architecture adheres to clean separation of concerns: routes delegate to ChatController, AI provider orchestration and streaming generators reside in ChatbotService, relational persistence is handled by Eloquent Models (Conversation & Message), and the interface is powered by a high-performance, single-file Blade view using Tailwind CSS, Marked.js, and Lucide icons.

2. Project Directory Structure & Responsibility Map

Below is the complete file breakdown showing exactly which file handles each specific responsibility in the codebase:

File / Directory Path	Core Component	Description & Specific Role
`app/Services/ChatbotService.php`	AI Service Provider	Connects directly to Google Gemini REST endpoint. Manages token streaming via PHP Generator (yield), model normalization, dual-header authentication (Authorization: Bearer + x-goog-api-key), automated 503/429 demand fallback, and shields users from technical errors.
`app/Http/Controllers/ChatController.php`	HTTP & Stream Controller	Orchestrates all web requests. Handles live SSE streaming in stream(), file uploads to public/uploads/ in upload(), conversation CRUD endpoints, system personas, model definition catalog, and MySQL message assembly.
`app/Models/Conversation.php`	Eloquent Model	Represents a chat thread in MySQL. Stores id, title, model, persona, temperature, and session_id. Defines hasMany(Message::class) relationship.
`app/Models/Message.php`	Eloquent Model	Represents individual conversation turns (user or assistant). Contains role, content, model, and a native JSON cast for attachments array.

`resources/views/chat.blade.php`	Frontend UI Workspace	Comprehensive single-page Blade interface. Features responsive collapsible drawer sidebar, Gemini model switcher popover, drag-and-drop file overlay, clipboard paste handler, audio STT / TTS, Markdown rendering, syntax highlighting with code copy buttons, and theme toggle.
`routes/web.php`	Routing Table	Maps root / and /chat to ChatController@index, POST /chat/stream for SSE streaming, POST /chat/upload for files, and RESTful conversation management routes.
`config/services.php`	Configuration	Defines credentials and connection metadata for Google Gemini (services.ai.key, services.ai.model, services.ai.url).
`.env`	Environment Config	Specifies runtime secrets: GEMINI_API_KEY, AI_PROVIDER=gemini, AI_MODEL=gemini-3.1-flash-lite, and MySQL database configuration.
`database/migrations/`	Database Migrations	Contains create_conversations_and_messages_tables.php and add_attachments_to_messages_table.php defining the relational schema in MySQL.
`public/uploads/`	Attachment Storage	Direct web-accessible destination directory for uploaded images (PNG, JPG, WEBP, GIF), documents (PDF, TXT, CSV), and code files.

3. Image Capabilities: Vision & AI Generation

A key architectural achievement of Nexus AI is dual image capability: it can both **understand user-uploaded images (Vision)** and **create new photorealistic images from text descriptions (Generation)**.

A. AI Image Generation (Text-to-Image)

What is used: The system leverages **Pollinations.ai (FLUX.1 / SDXL neural models)** integrated directly into the conversational pipeline without requiring external proprietary quotas or complex setup.

How it works:

- **Intent Detection & Prompt Refinement:** When a user asks to generate, draw, or visualize an image (e.g. *'generate an image of a cybernetic tiger in neon jungle'*), Gemini's system persona identifies the request and expands it into an exquisite, cinematic English prompt detailing lighting, texture, and aesthetic style.
- **Dynamic Markdown Embedding:** Gemini emits an optimized Markdown image URI: ![Description](https://image.pollinations.ai/prompt/{url_encoded_prompt}?width=1024&height;=1024&nologo;=true&seed;={rand}).
- **Seamless UI Rendering:** The frontend parses this Markdown via Marked.js and sanitizes it with DOMPurify. CSS class .prose-chat img renders the graphic with smooth rounded borders, subtle elevation drop-shadows, and interactive hover zoom.
- **Zero-Failure Fallback:** Even if the external API experiences a temporary timeout, ChatbotService::getGracefulFallbackReply() automatically extracts the user's prompt, cleans it, and returns the direct image markdown link instantly.

B. Multimodal Image Understanding (Image-to-Text Vision)

What is used: Google Gemini's native multimodal vision architecture (gemini-3.1-flash-lite & gemini-3.8-flash).

Workflow:

1. **Upload:** User uploads an image via the attach button (+), drags & drops files directly onto the chat window, or pastes a screenshot from clipboard (Ctrl+V).
2. **Storage:** ChatController@upload validates the MIME type and stores the file directly in public/uploads/, generating a web-accessible relative URL.
3. **Base64 Conversion:** For local development where external AI servers cannot probe localhost URLs, upload() encodes the image bytes into a Base64 Data URI (...).
4. **Gemini Vision Payload:** In ChatController@stream, image parts are formatted as standard multimodal parts: ['type' => 'image_url', 'image_url' => ['url' => $base64]] and delivered to Google Gemini for deep visual comprehension, OCR, diagram breakdown, or code reconstruction.

4. Google Gemini Models & Streaming Engine

Nexus AI is engineered with a real ChatGPT-style popover model switcher allowing users to toggle between different Google Gemini models based on task complexity:

Model Name	ID in Code	Strengths & Usage Profile	Speed
Gemini 3.1 Flash	`gemini-3.1-flash-lite`	**Default Model.** Ultra-fast multimodal workhorse. Instant streaming responses for everyday queries, chat, coding, and vision analysis.	Fastest
Gemini 3.5 Flash	`gemini-3.5-flash-lite`	High-throughput model with 1M token context. Excellent for massive documents, long codebases, and bulk file processing.	Fast
Gemini 4 Argon	`gemini-4-argon`	Maximum reasoning flagship model. Specialized in deep mathematical proofs, multi-step system architecture, and complex logic.	Thoughtful
Gemini 3.8 Flash	`gemini-3.8-flash`	Balanced multimodal intelligence for advanced agentic workflows and interactive reasoning.	Balanced

5. Enterprise Error Shield Architecture

A critical requirement of Nexus AI is that **no end-user is ever exposed to technical errors, API keys, HTTP status codes, or configuration advice**.

- **Server-Side Logging:** When an API call fails (HTTP 401, 404, 429, 503, or network timeout), the full technical diagnostic (status code, endpoint URL, error payload) is strictly logged to storage/logs/laravel.log for administrative diagnosis.
- **High-Availability Fallback:** If a specialized model (e.g. gemini-4-argon) experiences a temporary demand spike (503/429), ChatbotService automatically switches to gemini-3.1-flash-lite behind the scenes to complete the response.
- **Polite User Shield:** If all remote connectivity is severed, the user is presented only with a courteous message: *'I apologize, but I am currently experiencing a temporary connection issue. Please try again in a few moments.'*

6. MySQL Relational Database Schema

All conversations, individual messages, and attachments persist in MySQL (database: CHATBOT):

- **conversations Table:** id (BIGINT PK), user_id (nullable FK), session_id (VARCHAR UUID for guests), title (VARCHAR 255), model (VARCHAR 64), persona (VARCHAR 64), temperature (DECIMAL 3,2), created_at, updated_at.
- **messages Table:** id (BIGINT PK), conversation_id (FK), role (ENUM: 'user', 'assistant', 'system'), content (LONGTEXT), model (VARCHAR 64), attachments (JSON nullable: stores filename, size, URL, and metadata), created_at, updated_at.

7. Quick Start & Execution Commands

To run and verify Nexus AI in local development:

Command	Purpose / Description
`php artisan serve --port=8000`	Launches Laravel HTTP development server at **http://127.0.0.1:8000**.
`php artisan config:clear`	Flushes cached configuration after updating .env or service keys.
`php artisan migrate`	Executes database migrations creating conversations and messages tables.